Sets

A set is a gathering together into a whole of definite, distinct objects of our perception and of our thought which are called elements of the set.

The elements or members of a set can be anything: numbers, people, letters of the alphabet, other sets, and so on. Sets are conventionally denoted with capital letters.

Note: A set should be well defined and distinct.

Examples

(1) A = {tiger, lion, puma, cheetah, leopard, cougar, ocelot} (this is a set of large species of cats).

(2) A = {a, b, c, ..., z} (this is a set consisting of the lowercase letters of the alphabet)

(3) A = {-1, -2, -3, ...} (this is a set of the negative numbers)

In all above examples each element of the sets is distinct and well defined.

Operations on sets

Union

Two sets can be "added" together. The *union* of *A* and *B*, denoted by $\boxed{A \cup B}$, is the set of all things which are members of either *A* or *B*.

Examples:

- {1, 2} ∪ {red, white} ={1, 2, red, white}.
- {1, 2, green} ∪ {red, white, green} ={1, 2, red, white, green}.
- {1, 2} ∪ {1, 2} = {1, 2}.

Intersection

A new set can also be constructed by determining which members two sets have "in common".

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The *intersection of A and B*, denoted by $\boxed{A \cap B}$, is the set of all things which are members of both A and B. If $A \cap B = \emptyset$, then A and B are said to be *disjoint*.

$\{\ \}$, ϕ

ϕ $\emptyset$

Examples

$\{1, 2\} \cap \{\text{red}, \text{white}\} = \emptyset$.

$\{1, 2, \text{green}\} \cap \{\text{red}, \text{white}, \text{green}\} = \{\text{green}\}$.

$\{1, 2\} \cap \{1, 2\} = \{1, 2\}$.

Compliments

Two sets can also be "subtracted". The *relative complement* of B in A (also called the set-theoretic difference of A and B),

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denoted by $A \setminus B$ (or $A - B$), is the set of all elements which are members of A but not members of B.

Note: That it is valid to "subtract" members of a set that are not in the set, such as removing the element *green* from the set {1, 2, 3}; doing so has no effect.

Example:

- {1, 2} \ {red, white} = {1, 2}.
- {1, 2, green} \ {red, white, green} = {1, 2}. = $A \setminus B = A - B$.
- {1, 2} \ {1, 2} = ∅.
- If U is the set of integers, E is the set of even integers, and O is the set of odd integers, then $E' = O$.

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Cartesian products

A new set can be constructed by associating every element of one set with every element of another set. The *Cartesian product of two sets A and B*, denoted by $A \times B$ is the set of all Ordered pairs (a, b) such that a is a member of A and b is a member of B.

Examples

- {1, 2} × {red, white} = {(1, red), (1, white), (2, red), (2, white)}.
- {1, 2, green} × {red, white, green} = {(1, red), (1, white), (1, green), (2, red), (2, white), (2, green), (green, red), (green, white), (green, green)}.
- {1, 2} × {1, 2} = {(1, 1), (1, 2), (2, 1), (2, 2)}.

Memberships

The key relation between sets is *membership* when one set is an element of another. If a is a member of B, this is denoted $a \in B$, while if c is not a member of B then $c \notin B$. For example, With respect to the sets $A = \{1,2,3,4\}$ and $B = \{\text{blue, white, red}\}$, $4 \in A$ and green $\notin B$.

- Universal Sets: The universal set is the set of all things relevant to a given discussion and is designated by the symbol U. *i.e. it contains every set.*
- Subsets: If every member of set A is also a member of set B, then A is said to be a *subset* of B, written $A \subseteq B$ (also pronounced *A is contained in B*). The relationship between sets established by $\subseteq$ is called *inclusion or containment*.

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- Super Set: if we can write $B \supseteq A$, read as *B is a superset of A*, *B includes A*, or *B contains A*.
- Proper Subset: If A is a subset of, but not equal to, B, then A is called a *proper subset* of B, written $A \subset B$ (*A is a proper subset of B*) or $B \supset A$ (*B is a proper superset of A*).

Examples

- The set of all men is a proper subset of the set of all people.
- $\{1, 3\} \subset \{1, 2, 3, 4\}$.
- $\{1, 2, 3, 4\} \subseteq \{1, 2, 3, 4\}$.

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An obvious but useful identity, which can often be used to show that two seemingly different sets are equal:

$A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Power set: The power set of a set A is the set containing all possible subsets of A including the empty subset. It contains 2^n elements where n is the number of elements in A. It is typically denoted by $P(A)$ or 2^A. For example, the power set of the set $A=\{a,b,c,d\}$ is the set

$P(A)=\{\{\}, \{a\}, \{b\}, \{c\}, \{d\}, \{a,b\}, \{a,c\}, \{a,d\}, \{b,c\}, \{b,d\}, \{c,d\}, \{a,b,c\}, \{a,b,d\}, \{a,c,d\}, \{b,c,d\}, \{a,b,c,d\}\}$.

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A

Subset Relationships

$A = \{x \mid x \text{ is a positive integer} \leq 8\}$

set A contains: 1, 2, 3, 4, 5, 6, 7, 8

$B = \{x \mid x \text{ is a positive even integer} < 10\}$

set B contains: 2, 4, 6, 8

$C = \{2, 4, 6, 8, 10\}$

set C contains: 2, 4, 6, 8, 10

The universal set $U = \{1,2,3,4,5,6,7,8,9,10\}$.

Subset Relationships

$A \subseteq A$	$A \not\subset B$	$A \not\subset C$
$B \subset A$	$B \subseteq B$	$B \subset C$
$C \not\subset A$	$C \not\subset B$	$C \subseteq C$

Reflexive Relation: A Relation 'R' on a set 'A' is said to be Reflexive if $(x, x) \in R \ \forall \ x \in A$

$A = m$

$B = n$

$A \times B = \underline{m \times n}$

$A = \{1, 2, 3\}$ $\{1, 2, 3\}$

$$A \times A = \begin{Bmatrix} (1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3) \end{Bmatrix}$$

$R_1 = \{(1,1)(2,2)(3,3)\}$

$R_2 = \{(1,1)(2,2)\}$

$R_3 = \{\underline{(1,1)(2,2),(3,3)}(1,2)\}$

Examples: $R_1 = \{(x,y) \,/\, x-y \text{ is a integer}\}$ ✓

$R_2 = \{(x,y) \,/\, x-y \text{ is an odd no.}\}$

Check which of the above is Reflexive Relation on a set of Natural no.?

$(2, 3)$ $(2,2)$ $(4,4)$

How many Reflexive Relations?

$\{1,2\}$ then $A\times A = \left\{\begin{matrix}(1,1) & (1,2) \\ (2,1) & (2,2)\end{matrix}\right\}$ $2^2 = 4$. $1\times2\times2\times1$ 4.

$R_1 = \{(1,1)(2,2)\}$

$\{(1,1)(2,2)(1,2)\}$

$\{(1,1)(2,2)(2,1)\}$

$\{(1,1)(2,2)(1,2)(2,1)\}$

$\{1,2,3\}$ then $A\times A = \left\{\begin{matrix}(1,1) & (1,2) & (1,3) \\ (2,1) & (2,2) & (2,3) \\ (3,1) & (3,2) & (3,3)\end{matrix}\right\}$ $2^6 = 64$

$\{1,2,3,4\}$ then $\underline{A\times A} = \left\{\begin{matrix}(1,1) & (1,2) & (1,3) & (1,4) \\ (2,1) & (2,2) & (2,3) & (2,4) \\ (3,1) & (3,2) & (3,3) & (3,4) \\ (4,1) & (4,2) & (4,3) & (4,4)\end{matrix}\right\}$ $2^{12} = 4096$.

Total R − Ref.

$\underline{2^{n^2} - 2^{n^2-n}}$

$A = n$

$A\times A = n^2$

Diagonal $= n$

ND $\underline{n^2-n}$

Smallest $= n$

Largest $= n^2$

$\underline{2^{ND}} = 2^{n^2-n}$